

Power Unlimited

Wireless power, using a loaf-shaped device to power your entire house, mechanical strain generating electricity; the future of power generation has endless possibilities

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Piezo Electric Generators

In Israel, engineers have conducted successful experiments by embedding small stretches of the road with Piezo Electric Generators, which can convert mechanical strain into electrical current. It is possible that in one or two decades electricity generated from human movement will be used to fulfil the power needs of crowded places like railway stations, airports, movie theatres, amusement parks, etc.



Bloom Box

Bloom Box is based on the fuel cell technology and uses liquid or gaseous hydrocarbons (such as gasoline, diesel or propane) to generate electricity. The device is going to be of the size of loaf of bread and yet it will be able to power an entire house. The Bloom Box uses thin ceramic plates, which serve as the cathode and anode. Oxygen is fed into one side of the cell, while fuel is supplied to the other side, this leads to a chemical reaction and electricity gets generated. By 2025, advanced models of Bloom Box could be available.

For Monica Dutta this had been a demanding day, but nothing that she could not wash away in a warm bubble bath. The lights came on automatically the moment she stepped in the bathroom. The tub was brimful of water at just the right temperature. Sure, it took a lot of electricity to power such facilities, but in year 2025, electricity was no longer a problem. Every house had a range of options to power all kinds of devices and services. Monica had gone in for Wireless Power System to take care of much of her electricity needs. The systems in her house were capable of drawing power wirelessly from a remote microwave transmitter, owned by the local power utility company. But it was not as if all the electricity that the house needed came from the transmitter; the house was also capable of generating electricity on its own through the system of piezo electricity.

The floors were embedded with Piezo Electric Generators to convert every bit of mechanical strain caused by movement into electricity, which got stored in super efficient Nano-tech Batteries, remotely connected to every system in the house. In fact, in year 2025 many of the busy arterial roads had been fitted with piezo electric generators, which converted the strain caused by vehicular movement and footfalls into energy that got fed into the grid.

After emerging from the bathroom, Monica went to her sitting room, and ensconced herself on the sofa. "TV," she said. A paper-thin screen immediately dropped down from the roof and stationed itself at optimum distance from her eyes. Through the movement of her eyelids she browsed the channel that she wanted to watch. Whenever she moved her head, the screen automatically adjusted its position to remain in front of her.

Monica's house had an efficient AI to take care of many of the routine chores like security, cleaning, cooking, maintenance of the temperature at optimal level, management of music, entertainment systems, as well as the workstation, and much else. The AI drew its power wirelessly from the grid, and also from the nano-tech batteries. As a form of extra security, Monica had the AI fitted with Bloom Box, which could keep the emergency functions in the house running even if the power supply in the grid and in the nano-batteries failed.

Even the car that she drove drew its power from the grid. Yesterday she was zipping down the super-expressway, bound for an important meeting, when a beep alerted her to the fact that her car was about to run out of power. Without batting an eyelid she pressed a button on the dashboard and the car's computer system automatically got connected to the microwave transmitter located close to the expressway. The battery got charged while the vehicle continued to cruise down the highway. Solar power was also very common, not only for individual homes and offices, but also for vehicles. She had another car that was capable running solely on solar power.



Nano-tech batteries

Nano-tech batteries use billions of tiny, virtually identical nano-structures to perform the task of receiving, storing and delivering substantial electricity. Since nano-structures are capable of interacting at high speeds, these batteries can be recharged very fast and have larger charging cycles. Researchers working at the Maryland NanoCenter, in University of Maryland, have already developed the prototype for a nano-tech battery for daily use.



Solar Energy

The technology by which we can harness solar energy has been improving by leaps and bounds. Scientists are currently engaged in improving the scope and efficiency of solar panels by employing principles of nano-technology.

It is possible that in future we might have solar panels that also incorporate a nano-battery, so that the power that gets generated will be directly stored in the panel itself before being transferred to the home, office or the grid. Some scientists are also exploring the possibility of using an array of mirrors to focus the sun's rays to a small area, which then becomes very hot and is used to drive steam turbines that will generate electricity.